

**THIRD SEMESTER (CUCBCSS-UG) DEGREE EXAMINATION
NOVEMBER 2021**

Mathematics

ME 3C 03—MATHEMATICAL ECONOMICS

(2014–2018 Admissions)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes**Total No. of Questions : 20****Maximum : 20 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

ME 3C 03—MATHEMATICAL ECONOMICS
(Multiple Choice Questions for SDE Candidates)

1. In the production function $Q = AK^\alpha L^\beta$, the variable L denotes :
(A) Leisure. (B) Less.
(C) Labour. (D) Loss.
2. Curve that is also known as equal product curve is :
(A) Indifference curve. (B) Isoquants.
(C) Demand Curve. (D) None of the above.
3. A homogeneous production function with degree one corresponds :
(A) Constant returns. (B) Diminishing returns.
(C) Increasing returns. (D) Negative returns.
4. Given $\frac{\partial Q}{\partial x_1}$, where x_1 is input represents :
(A) AP_{x_1} . (B) MP_{x_1} .
(C) MRTS. (D) MRS.
5. When the total product is maximum, marginal product will be ?
(A) Minimum. (B) Zero.
(C) Maximum. (D) Negative.
6. When demand for a good is give by $Q = 40 - P$, the maximum amount that would be demanded at nil price is :
(A) 1. (B) 0.
(C) 40. (D) 400.
7. Combinations of two inputs resulting in equal total output is :
(A) Isoquant. (B) Isocost.
(C) Indifference curve. (D) Priceline.

8. Which of the following shows constant returns to scale ?

- (A) Cobb Douglas production function. (B) CES production function.
(C) Both (A) & (B). (D) None of the above.

9. $\frac{Q}{x_2}$, where x_2 denotes inputs corresponds to :

- (A) MP_{x_2} . (B) MP_{x_1} .
(C) AP_{x_2} . (D) AP_{x_1} .

10. IN the CES production function $Q[\delta C^{-\sigma} + (1 - \delta)N^{-\sigma}]^{\frac{-1}{1-\sigma}}$ the term C denotes :

- (A) Constant. (B) Output.
(C) Capital. (D) Population.

11. Which of the following is called product exhaustion theorem ?

- (A) Eulers's theorem. (B) Cobb-Douglas function.
(C) CES. (D) Translog.

12. Demand function $Q = f(P)$ if point elasticity $\epsilon = -1$ for all $P > 0$ will be :

- (A) CP. (B) $\frac{c}{p}$.
(C) P. (D) None of the above.

13. What is the order of $\left(\frac{d^2y}{dt^2}\right)^7 + \left(\frac{d^3y}{dt^3}\right)^5 = 100y$:

- (A) 2. (B) 3.
(C) 5. (D) 7.

14. A line for linear equation should begin from :

- (A) Origin. (B) X-axis.
(C) Y-axis. (D) Any of the above.

15. Euler's theorem is valid only if factors are paid reward on the basis of value of :

- (A) Average product.
- (B) Total Product.
- (C) Marginal product.
- (D) None of the above.

16. The order of $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 = 25x$ is :

- (A) First.
- (B) Second.
- (C) Third.
- (D) None of the above.

17. The value of Rs. 100 at 10% interest for two years :

- (A) 110.
- (B) 111.
- (C) 121.
- (D) 130.

18. Warranted growth rate in the Harved model is :

- (A) SYt .
- (B) $\frac{s}{a-s}$.
- (C) $\frac{a}{a-s}$.
- (D) $\frac{a-s}{a}$.

19. Functional relationship between input and output is called :

- (A) Isoquants.
- (B) Isocost.
- (C) Input function.
- (D) Production function.

20. What is the order of $\frac{d^3Y}{dx^3} + (x^2Y)\frac{d^2Y}{dx^2} - 4Y^4 = 0$:

- (A) First.
- (B) Second.
- (C) Third.
- (D) Fourth.

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NOVEMBER 2021**

Mathematics

ME 3C 03—MATHEMATICAL ECONOMICS

(2014–2018 Admissions)

Time : Three Hours

Maximum : 80 Marks

Part A

*Answer all the **twelve** questions.
Each question carries 1 mark.*

1. Write the order and degree of the differential equation $\left(\frac{d^2y}{dt^2}\right)^5 + \left(\frac{d^3y}{dt^3}\right)^4 - 36y = 0$.
2. Solve $\frac{dy}{dt} = y + 1$.
3. Write the general solution of the difference equation $y_t + \frac{1}{4}y_{t-1} = 0$.
4. Mathematically, how will you represent a production function.
5. Define the term Marginal product.
6. Define Marginal rate of technical substitution.
7. State Euler's theorem.
8. Check the returns to scale of the production function $Q = A K^{0.3} L^{0.7}$.
9. For the production function $P = AL^\alpha K^\beta$, prove that labour L and capital K are essential factors of production.
10. Define the term Pay-back period.
11. Write any two advantages of ARR method.
12. Write a short note on Decision tree approach.

(12 × 1 = 12 marks)

Turn over

Part B

*Answer any **six** questions.
Each question carries 3 marks.*

13. Solve the differential equation $\frac{dy}{dt} = \frac{y^2}{t^2}$.
14. Find the demand function $Q = f(p)$ if point elasticity is $\frac{-(5p + 2p^2)}{Q}$ and $Q = 500$ when $P = 10$.
15. Solve the difference equation $y_t = y_{t-1} - 25$ given $y_0 = 40$.
16. Explain why an Isoquant slopes downward from left to right.
17. Define the term Producers equilibrium.
18. Write the general form of CES production function.
19. Given the production function $Q = AK^\alpha L^\beta$, find out Marginal productivity of capital (K) and labour (L).
20. A project of Rs. 20 lakhs yielded annually a profit of Rs. 4 lakhs after depreciation at 10% and is subject to income tax at 40%. Calculate pay-back period of this project.
21. Write any three advantages of NPV method.

(6 × 3 = 18 marks)

Part C

*Answer any **six** questions.
Each question carries 5 marks.*

22. Show that $(12y + 7t + 6) dy + (7y + 4t - 9) dt = 0$ is exact and hence solve.
23. For the following data find the level of income y_t for any period and warranted rate of growth :
 $I_t = 4.2(y_t - y_{t-1})$; $s_t = 0.2 y_t$; $y_0 = 5600$.
24. Show that the marginal product is always equal to the average product when average product is maximum.
25. If $P = f(A, B)$ is a linear homogeneous production function, the Euler's theorem $P = A \frac{\partial P}{\partial A} + B \frac{\partial P}{\partial B}$.
26. If the profit rate (price of capital) remains unaltered the ratio of the amount of capital employed per unit of labour shifts from 10 : 10 to 12 : 11, given that the rise in wages is 25%. Determine the elasticity of substitution.

27. Prove that the expression path of the Cobb-Douglas production function is linear and passes through the origin.
28. Prove that for the C.E.S. production function the marginal product is greater than zero.
29. Compute the comparative profitability of the two proposals using the method of ARR :

Particulars		Investment Proposal for	
		Machine A	Machine B
1	Initial cost (Rs.)	2,24,000	60,000
2	Estimated life (years)	5.5	8
3	Estimated sales (Rs.)	1,50,000	1,50,000
4	Costs (Rs.)		
5	Material	50,000	50,000
6	Labour	1,20,000	60,000
7	Overheads	24,000	20,000

30. Write a note on simulation approach.

(6 × 5 = 30 marks)

Part D

Answer any two questions.

31. For the data given below, determine (i) The market price P_t in any time period ; (ii) The equilibrium price, P_e ; (iii) The stability of time path :

$$Q_{dt} = 160 - 0.8 P_t ; Q_{st} = -20 + 0.4 P_{t-1} ; P_0 = 153.$$

32. Find the elasticity of substitution for the production function $P = AL^\alpha K^\beta$ with $\alpha + \beta = 1$.
33. Derive producer's equilibrium condition if the production function is $P = f(A, B)$ and the prices of the factors A and B are P_A and P_B respectively.

(2 × 10 = 20 marks)